SALTDRAG · STEM CHECK

# SaltDrag STEM check

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## DOCUMENT CONTROL

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| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/saltdrag | | |
| Source | docs/01b-stem-check.md @ c7f0938 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.2 | 2026-10-25 | Amish Chadha | S24 note: sets per pan decided (D25); no verdict changed |
| 0.1 | 2026-10-24 | Amish Chadha | First STEM check (kit 1.8 upgrade of a TRL 3 design); four detail failures fixed under Amish's standing instruction of 2026-10-23, five more after the independent review (SDG-DEC-001 D17 to D24); S7 reopened by the review |

# SaltDrag STEM check

Verdict: the working principle (two people wind a geared drum on dry ground and the rope draws a rake across the pan) sits within known science, and so do the principles of options B and C. The first check found four details that failed on the TRL 3 design of 2026-10-03: each cranker worked at 84 W against a sustained 75 W (S15), the 4.5 mm drum barrel kept a margin of only about 1 under three layers of rope at twice the rated pull (S7), three bolted joints could not be assembled as drawn (S26), and the assumption that several families share one set did not survive a throughput check (S24). The independent review then refuted five more verdicts: the brake would not hold with a wet lining (S9), one chain can take most of the pull (S11), the 25 mm crank shaft was short of its margin once C45's yield is taken for its size and of fatigue life in salt air (S5, S17), the gear and ratchet keys were overloaded (S22), and the rake would pitch onto its blade with the bridles at mid-height (S29). Each was fixed under Amish's standing instruction of 2026-10-23 and is recorded in SDG-DEC-001 (D17 to D24); the rows below are on the fixed design. Of the 31 rows for option A, **25 are within known science and 6 are plausible, to confirm at TRL 4 with field data, a test or a datasheet**: the drag per metre of rake (S1, on which S2, S15, S23 and S30 depend), the drum barrel's multi-layer rope pressure (S7, reopened by the review), the anchor holding in the soil behind the bund (S12), safe working hours in the harvest heat (S16), 316 stainless in brine (S19) and the rope's life over the roller and on the bed (S27). None can be closed by calculation, so `stem_check` is **open**. Method and rules: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which reads the results of SDG-CAL-001.

## Claims checked

Table 1. Claims, governing science and verdicts for option A, the current design (generated from `docs/04-calcs/stem_results.csv`).

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| S1 | A 3 m rake with a full heap of about 231 kg needs about 2.75 kN to drag (0.9 kN per metre) | Physics | Mechanics and statics | Coulomb sliding of the heap (friction 0.70, tan of the 35 degree repose angle) plus an assumed 300 N/m to loosen the harvest layer and skid friction 0.30; granular drag in front of a blade (soil-tool mechanics) | heap 1589 N + layer 900 N + skids 259 N = 2748 N; the layer term is an estimate, not a measurement (SDG-CAL-001 [A7] rests on the assumed 300 N/m; measured on the partner's bed (SDG-DEC-001, to confirm 7)) | Plausible, to confirm at TRL 3 | Open |
| S2 | At the design drag of S1, two people at 89 N each on 330 mm cranks wind the rake | Physics | Mechanics and statics | Torque balance through a 6 to 1 spur stage, 94 % gear and bearing efficiency, rope on its third layer; if the drag is 1.5 times the estimate the pull rises 1.42 times and the skids are raised or the rake shortened (SDG-CAL-001 [A10]) | 89.4 N <= 120 N | Within known science | SDG-CAL-001 [B3]; PR-002 |
| S3 | Two people at 120 N give at least 3 kN of rope pull, even on a full drum | Physics | Mechanics and statics | Torque balance as S2 on the fourth layer (largest radius) | 3.68 kN >= 3 kN | Within known science | SDG-CAL-001 [B5]; PR-003 |
| S4 | The 40 mm drum shaft keeps a margin of 2 at twice the rated pull (6 kN) | Physics | Strength of materials | Bending (rope load and the gear's reaction, both planes) plus torsion, von Mises; C45+QT yield 430 MPa, 16 to 40 mm (EN 10083-2) | 158.8 MPa <= 215 MPa | Within known science | SDG-CAL-001 [C1]; PR-003 |
| S5 | The crank shaft at the bearing next to the overhung pinion keeps | Physics | Strength of materials | Bending from the full tooth force plus torsion, von Mises, 30 mm; C45+QT yield 430 MPa | 95.7 MPa <= 215 MPa | Within known science | SDG-CAL-001 [C2]; PR-003 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | a margin of 2 at twice the rated pull (6 kN) | | | | | | |
| S6 | The module 4 pinion teeth keeps a margin of 2 at twice the rated pull (6 kN) | Physics | Strength of materials | Lewis bending, form factor 0.277 for 14 teeth; C45+N yield 305 MPa above 16 mm | 138.6 MPa <= 152.5 MPa | Within known science | SDG-CAL-001 [C3]; PR-003 |
| S7 | The drum barrel is not crushed by three layers of rope wound on at 6 kN, with a margin of 2 | Physics | Strength of materials | Hoop stress of a ring under rope pressure, sigma = k T / (t s); 168.3 x 6.3 YSt 310 tube, yield 310 MPa (IS 4923); the multi-layer factor k = 1.8 for three loaded layers is taken from wire-rope practice and is not sourced for polyester (independent review); the margin of 2 holds for k up to 1.95; flange thrust not yet checked | 143 MPa against 155 MPa at k = 1.8 (margin 2.17); the 4.5 mm pipe it replaces failed at about 200 MPa (sourced layer factor and a CalRig proof load with three layers wound on (TRL 4); SDG-DEC-001 D18) | Plausible, to confirm at TRL 3 | Open |
| S8 | The pawl boss welded to the cheek keeps a margin of 2 at twice the rated pull (6 kN) | Physics | Strength of materials | Cantilever bending of the 40 mm boss; S275 yield 275 MPa (EN 10025-2) | 58.8 MPa <= 137.5 MPa | Within known science | SDG-CAL-001 [C5]; PR-003 |
| S9 | A hand force of 150 N or less on the band brake holds the rated pull, with the lining wet | Physics | Mechanics and statics | Band brake (capstan) equation T1/T2 = exp(mu theta), mu 0.20 for a woven lining wet with brine (0.30 dry), 247 degrees of wrap, differential | 134.1 N <= 150 N | Within known science | SDG-CAL-001 [C7]; PR-011 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | | | | lever 600 mm with a 50 mm inner arm | | | |
| S10 | The rope's minimum breaking strength after splices is at least 5 times the rated pull | Engineering and technology | Standards practice | Working-load factor of 5 to 12 for fibre rope in general use (rope makers' practice); 12 mm polyester double braid 27 kN class, splices 85 % | 7.65 MBS / rated pull >= 5 MBS / rated pull | Within known science | SDG-CAL-001 [C8]; PR-010 |
| S11 | The more loaded anchor chain keeps a margin of 2 on its working load limit at 6 kN | Engineering and technology | Standards practice | Chain tension from the 23 degree chain line with 70 % of the pull on one chain (unequal tension, independent review); 10 mm grade 30 chain working load limit 11.8 kN (2,650 lb, catalogue class, to confirm on purchase) | 2.59 WLL / tension >= 2 WLL / tension | Within known science | SDG-CAL-001 [C9b]; PR-003 |
| S12 | Each 150 mm helical anchor holds at least 1.5 times its share of the rated pull in the ground behind the bund | Physics | Mechanics and statics | Deep helix bearing Q = 9 s_u A in firm clay, s_u 50 kPa assumed; the rod is screwed in at 45 degrees and the chain arrives at 23 degrees, so the load is 22 degrees off the anchor's axis; at a helix depth of about 0.6 m the bearing factor may be below 9, and saline silty soil can be softer | 7.95 kN estimated against a proof pull of 3.42 kN (1.5 times a 70 % share); 1.74 times the 6 kN chain force at a 70 % share; soil unknown (SDG-CAL-001 [C10]; site pull test before first use (SDG-DEC-001, to confirm 6)) | Plausible, to confirm at TRL 3 | Open |
| S13 | The anchored capstan does not tip, even at twice the rated pull | Physics | Mechanics and statics | Moments about the front tips of the runners: rope at 677 mm against weight, chain tension and chain line, both chains taut (rated pull | 1.46 resisting / tipping >= 1 resisting / tipping | Within known science | SDG-CAL-001 [C12], [C12b] |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | | | | 1.56); a slack chain is a hazard controlled by the turnbuckles and the set-up check (SDG-RSK-001 F11) | | | |
| S14 | The 3 m FRP rake beam keeps a margin of 2 at twice the rated pull (6 kN) | Physics | Strength of materials | Simply supported beam under the heap and layer load; pultruded FRP flexural strength 200 MPa (lower bound) | 28 MPa <= 100 MPa | Within known science | SDG-CAL-001 [C13]; PR-003 |
| S15 | At the design drag, each of two crankers works within a sustained rate, even at the end of a pass | Biology | Ergonomics | Rope power over efficiency, shared by two, with the rope on its third layer at 23 turns a minute (mean over a pass 64 W; pull duty about 45 %); sustained hand-cranking by an average adult about 75 W (.kit/ stem_check.py limit) | 71.1 W <= 75 W | Within known science | SDG-DEC-001 D17; SDG-CAL-001 [B7b]; PR-018 |
| S16 | Winding at about 70 W is safe in the harvest heat with work planned for the cooler hours | Biology | Human physiology and biomechanics | Arm cranking is about 15 % efficient, so 64 to 71 W of external work is a metabolic rate of about 450 to 500 W, high work; ISO 7243 sets a reference WBGT of about 26 C for acclimatised workers at high work (to confirm against the 2017 edition); afternoon WBGT in the Little Rann in March and April is likely to exceed it, so 6 working hours a day | safe in the early morning and evening only; a work and rest schedule is needed (work hours, shade and water agreed with the partner (PRB co-design checklist; RSK H6)) | Plausible, to confirm at TRL 3 | Open |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | | | | (S24) may be optimistic | | | |
| S17 | The crank shaft survives a season of rotating bending at the rated pull | Physics | Strength of materials | Goodman, fully reversed bending from the tooth force and steady torque at the 25 mm step (notch factor 1.6), C45+QT (UTS 650 MPa), surface and size factors, endurance halved for bare steel in salt air (corrosion fatigue, estimate) | 1.59 safety factor >= 1.5 safety factor | Within known science | SDG-DEC-001 D24; SDG-CAL-001 [C16], [C16b] |
| S18 | Hot-dip galvanised steel on the dry ground lasts more than one season in salt-laden air with a daily rinse | Chemistry | Electrochemistry and batteries | Zinc corrosion rate about 4.2 to 8.4 um a year in a very high (C5) atmosphere and up to about 25 um a year in an extreme (CX) one (ISO 9223); minimum mean hot-dip coating 85 um on steel over 6 mm and 55 um on 1.5 to 3 mm (ISO 1461); cut edges and drilled holes are galvanised after fabrication | at 25 um a year (brine-wetted, CX): 3.4 years on the 6 mm plate, 2.2 years on the 3 mm runners and tubes; more than one season | Within known science | BOM 1 to 3, 10 to 12; PR-007 |
| S19 | 316 stainless fittings on the rake survive a season wetted by near-saturated brine | Chemistry | Electrochemistry and batteries | Type 316 suffers pitting and crevice corrosion in warm chloride brines, worst in crevices under bolt heads and washers (Nickel Institute guidance on stainless steel in seawater); daily freshwater rinse lowers the risk | no calculation possible; crevice corrosion of the blade and hanger bolts is the likely failure (season trial with inspection (PR-007, TRL 5); duplex 2205 bolts as the fallback (RSK F9)) | Plausible, to confirm at TRL 3 | Open |
| S20 | UHMW-PE, pultruded | Chemistry | | Polyolefins and glass-polyester or | | | |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | FRP and polyester rope keep their strength in brine and sun for a season | | Materials and polymer chemistry | vinylester composites are inert to sodium chloride; UV degrades unstabilised UHMW-PE and bare FRP surfaces, so the BOM asks for UV-stabilised UHMW-PE and a surfacing veil on the FRP | chemically inert; UV handled by grade choice | Within known science | BOM 13, 14, 15, 18; PR-007 |
| S21 | No metal coating, paint, grease or FRP touches the salt | Chemistry | Environmental and atmospheric chemistry | The salt touches only the UHMW-PE blade and skids (food-contact grades exist), 316 stainless and the polyester rope; no zinc, paint, grease or FRP is in the pan below the blade top; the rope may shed fibres where it drags on the bed (to watch in trials) | material list by inspection of the model | Within known science | cad/src/model.py option A; BOM 14 to 16 |
| S22 | The gear and ratchet keys carry the drum torque at 6 kN with a margin of 2 | Physics | Strength of materials | Key bearing stress 2 T / (d (h/2) L), 12 x 8 keys on the 40 mm shaft: gear hub 44 mm (C45+QT, 430 MPa), ratchet with its welded hub 64 mm (C45+N, 305 MPa) | 2.08 margin >= 2 margin | Within known science | SDG-DEC-001 D24; SDG-CAL-001 [C17], [C18] |
| S23 | At the design drag, one set gathers about 532 kg of salt an hour with a crew of four | Mathematics | Orders of magnitude | One full heap per strip over pull, return and move times | claimed 532.124 kg/h, estimate 532 kg/h (ratio 1.00) | Within known science | SDG-CAL-001 [D8], [D9] |
| S24 | One set gathers about 195 t in a 61-day harvest, | Mathematics | Orders of magnitude | Rate x 61 days x 6 cooler hours; a family pan yields 500 to 600 t a year | claimed 194.757 t, estimate 201 t (ratio 0.97) | Within known science | SDG-DEC-001 D20; SDG-CAL-001 |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| | about a third of one family pan's yearly output (6 cooler hours a day; about 130 t at 4 h) | | | (Business Standard, 2009) | | | [D11], [D12] |
| S25 | Each drum stores the 45 m of rope a 30 m pan needs | Mathematics | Units and dimensional analysis | Turns per layer x layer circumference, four layers of 12 mm rope on a 168.3 mm barrel | 58.5 m >= 45 m | Within known science | SDG-CAL-001 [B8]; PR-012 |
| S26 | Every bolted joint can be assembled: washers and nuts clear their neighbours, holes keep at least 1.2 diameters of edge | Engineering and technology | Manufacturing tolerances | Edge distance e >= 1.2 d0 (EN 1993-1-8 practice); interference check of every part at detail level D3 (.kit/cad_check.py) | cross-tube bolts at 36 mm on 100 x 80 plates (14 mm edge, 1.27 d0), guard spacer holes 73 mm from the cheek edge, axle split pins outside the plates, gear and ratchet hubs clear of the bearings, guard and crank; cad/fidelity.json: no interference | Within known science | SDG-DEC-001 D19, D24; cad/fidelity.json |
| S27 | The rope lasts a season over the 100 mm roller and the 168 mm drum and dragging on the bed | Engineering and technology | Standards practice | Rope makers give a minimum sheave-to-rope diameter ratio of about 8 for polyester double braid (to confirm on the chosen rope's datasheet); a bend at D/d 8.3 costs about 15 % of strength (the working-load factor falls from 7.7 to about 6.5, still over 5); salt crystals abrade the rope where it lies on the bed on the light return | roller 8.3, drum 14.0 against about 8; abrasion life unknown (rope datasheet on purchase (SDG-DEC-001, to confirm 10); daily inspection (SDG-RSK-001 F1)) | Plausible, to confirm at TRL 3 | Open |

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| S28 | The induction-hardened pinion teeth resist pitting at the rated pull | Physics | Strength of materials | Hertz contact sigma_H = Z_E Z_H sqrt(K_A F_t (u + 1) / (b d u)), K_A 1.25 for hand cranking; allowable about 1,000 MPa for induction-hardened C45 teeth (estimate) | 1.49 margin >= 1.2 margin | Within known science | SDG-DEC-001 D24; SDG-CAL-001 [C3c], [C3d] |
| S29 | The rake does not pitch onto its blade edge under the full heap | Physics | Mechanics and statics | Moments about the skid tips: the pull at the bridle height (70 mm) against the heap at a third of the blade height, the layer at the edge and skid friction at the floor; restoring moment of the rake's weight about the skid tips | 9.81 restoring / pitching >= 1.5 restoring / pitching | Within known science | SDG-DEC-001 D23; SDG-CAL-001 [D13] |
| S30 | With two 24 kg ballast bags the blade stays down until about 3.5 m from the bund toe | Physics | Mechanics and statics | Vertical equilibrium of the rake under the rising rope: lift-off when T sin(theta) exceeds the rake's weight | claimed 3.5 m, estimate 3.5 m (ratio 1.00) | Within known science | SDG-CAL-001 [D4] |
| S31 | The skids press on the pan floor less than a worker standing on it does | Physics | Mechanics and statics | Contact pressure = weight / skid area; the pull's pitching moment is small with the bridles at the skid foot (S29), so the pressure stays near the static value; a 70 kg person on two feet (about 0.025 m2 each) presses 13.7 kPa | 10.7 kPa <= 13.7 kPa | Within known science | SDG-CAL-001 [D1]; PR-004 |

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. The kit's label "Plausible, to confirm at TRL 3" is kept for the six open rows; each needs a measurement, a test or a datasheet, so it closes at TRL 4. Row numbers follow the order the claims were raised; S17, S22, S28 and S29 were added after the independent review.

## Fixes made under the standing instruction

Table 2. Rows that failed on the 2026-10-03 design and the fixes made (SDG-DEC-001).

| Row | What failed | Fix | Result |
|---|---|---|---|
| S15 | At 30 crank turns a minute each of the two crankers works at 84 W on the loaded pull, above the 75 W an average adult sustains when cranking; the first fix, 25 turns a minute, still gave 77 W with the rope on its third layer (independent review) | D17: design cranking rate 23 turns a minute; the crank force at the design drag is unchanged (89 N) | 71 W at the end of a pass, 64 W on average; the pass takes 12.4 min instead of 9.5, and the estimated rate falls from 598 to 532 kg/h (150 to 133 kg per worker-hour) |
| S7 | The drum barrel, 168.3 x 4.5 mm pipe (yield about 210 MPa), carries the pressure of three layers of rope wound on at the drum: about 200 MPa at twice the rated pull with a multi-layer factor of 1.8, a margin of 1.05 where PR-003 asks for 2 | D18: barrel 168.3 x 6.3 mm YSt 310 tube (IS 4923; S355J2H under EN 10210), same outside diameter | 143 MPa, margin 2.17 at a factor of 1.8; the review showed the factor is not sourced for polyester, so the row stays open (Plausible) |
| S26 | Modelling every fastener at D3 found three joints that could not be assembled as drawn: the cross-tube washers sat 2 mm from the tube walls where the 5 mm fillet welds go; the side gear guard spacer holes were 2.5 mm from the cheek edge; the roller axle's clip holes fell inside the saddle plates | D19: cross-tube end plates 100 x 80 mm with the bolts 36 mm from the tube centre; guard spacers 150 mm ahead and behind and 120 mm below the drum axis; roller axle 272 mm with 6 mm split pins; bearing bolts fitted with their heads inside the cheeks | Every washer, nut and weld clears its neighbours; `cad/fidelity.json` reports no interference |
| S24 | The requirements assumed several families could share one set. One set gathers about 195 t in a 61-day harvest at 6 h a day, about 35 % of one pan's yearly 500 to 600 t | D20: the sharing assumption is withdrawn from PRS and MKT; one set serves one family pan; how a family gathers the rest went to Amish as open decision 2, decided on 2026-10-25: one set per pan stays the design case until the partner answers (SDG-DEC-001 D25) | The claim now stated is the one the numbers support |
| S9 | The brake held the rated pull with 92 N only with a dry lining (friction 0.30); wet with brine (0.20) it needs 179 N, over the 150 N of PR-011 (review) | D21: brake lever 600 mm (was 450); lining kept dry and free of grease | 134 N with a wet lining |
| S11 | The 8 mm chains kept a margin of 2.4 only if both share the pull equally; with 70 % on one chain the margin is 1.71 (review) | D22: 10 mm grade 30 chain (WLL 11.8 kN), rated forged turnbuckles and 1 t shackles; anchor proof pull raised to 3.42 kN | Margin 2.6 on the more loaded chain |
| S5, S17 | C45 normalised has a yield of 305 MPa above 16 mm, not 340: the 25 mm crank shaft fell to a margin of 1.89 at 6 kN, and with rotating bending in salt air its fatigue factor was below 1.5 (review) | D24: crank shaft C45+QT, stepped 30 mm through UCF206 bearings and 25 mm outboard; drum shaft C45+QT | Crank shaft margin 4.5 at the bearing, 3.9 at the step; fatigue factor 1.59 at the step with the endurance halved for corrosion |
| S22 | The 12 x 8 keys sat in a 30 mm gear face and a 20 mm ratchet wheel: bearing stress at 6 kN 303 and 455 MPa, at or above | D24: gear with a 44 mm hub in C45+QT; ratchet with a welded 44 mm hub, the drum shaft 26 mm longer | Key margins 2.08 and 2.15 |

| Row | What failed | Fix | Result |
|---|---|---|---|
| | yield (found while checking the review's list of unchecked parts) | | |
| S29 | The bridles pulled at mid-height (130 mm) while the heap and layer resist lower down: the pitching moment was about twice the rake's restoring moment, so the blade would tip onto the floor under a full heap (from the review's note on skid pressure) | D23: eye plates at the foot of the end caps, bridle apex at 70 mm | Restoring moment 9.8 times the pitching moment; the windrow moves from 3.3 to 3.5 m from the bund toe |

None of the fixes changes the working principle, the pitch or the user-facing scope: a hand capstan still winds a rake across the pan from the bund. All are recorded in SDG-DEC-001 as decided under Amish's standing instruction of 2026-10-23, with the options weighed and the cost, and he can reverse any of them.

## Fields not relevant

None. Every field is touched: physics (statics, strength of materials, friction), chemistry (corrosion of zinc and stainless steel in salt, polymer durability, contamination of the salt), biology (cranking power, heat stress), mathematics (orders of magnitude of rate and throughput, rope capacity) and engineering (rope practice, chain ratings, assembly tolerances).

## Concept options

Table 3. Option rows (first-order; numbers from `docs/04-calcs/stem_check.py`).

| # | Claim | Field | Branch | Governing law or source | First-order check | Verdict | Closed by |
|---|---|---|---|---|---|---|---|
| B1 | A team of 14 walking haulers draws a 1.5 m rake at 1.73 kN (option B) | Biology | Ergonomics | Sustained walking pull of about 130 N per person, the order of the maximum acceptable sustained pull forces for most adult women (Snook and Ciriello, 1991); rope pull from the drag model of S1 at half width | 14 x 130 N >= 1729 N; possible, but the crew is large | Plausible, to confirm at TRL 3 | Option not carried forward |
| B2 | A post on four 25 mm stakes holds the 90 degree rope turn (option B) | Physics | Mechanics and statics | Resultant of the turned rope, 1.41 times the pull, at 0.76 m height; lateral capacity of short stakes in clay | 2.44 kN at 0.76 m; stake capacity depends on the soil | Plausible, to confirm at TRL 3 | Option not carried forward |
| C1 | Two people at about 140 N on a 10 to 1 hand lever pull the 3 m rake (option C) | Physics | Mechanics and statics | Lever ratio 1052 / 81 mm, 85 % efficiency at pins and jaws, the rope pull of SDG-CAL-001 [A8] | 136 N per person <= 150 N per person | Within known science | Option not carried forward |
| C2 | Cam jaws grip polyester rope without wearing it out within a season (option C) | Engineering and technology | Standards practice | Toothed cam grips cut and glaze fibre rope under repeated slip; commercial fibre-rope pullers use smooth or grooved jaws and limit the load | rope wear under 600 strokes a pass is unknown | Plausible, to confirm at TRL 3 | Option not carried forward |

- Option A, geared drum capstans: the rows of Table 1. Within known science; six rows open for field data or a test.
- Option B, bund-walk haul: within known science. Walking haulers pull with their legs and body weight, but about 130 N each sustained, so a 1.5 m rake needs about 14 haulers (B1); the staked turning post depends on the soil (B2). Not failed; not recommended (SDG-OPT-001).
- Option C, lever rope puller: within known science. A 13 to 1 lever brings the effort down to about 136 N for each of two people (C1), but the rope moves about 52 mm a stroke and toothed cam jaws wear fibre rope (C2). Not failed; not recommended.

## Independent review

A second agent (run headless with `claude -p`), given only four files, this STEM check, the calculation note SDG-CAL-001, the product requirements SDG-PRS-001 and `docs/04-calcs/sizing.py`, and none of the author's reasoning, was asked on 2026-10-24 to refute the headline results and every "Within known science" verdict. It refuted three verdicts outright (S2 and the rows that inherit the assumed drag, S5 and S9), weakened most others, and confirmed the arithmetic of S3, S4, S8, S10, S11, S18, S25 and Section A. Its findings and what changed:

| Row | Finding | What changed |
|---|---|---|
| S2, S15, S23, S24, S30 and the CAL headline | All inherit the assumed drag of S1; the design drag equals the rated pull by construction, and S23 and S30 only test the arithmetic | Claims now say "at the design drag"; drag sensitivity added to CAL ([A9] to [A11]: 1.5 times the drag needs 1.42 times the pull); S1 stays open |
| S15 | Power taken at the mean rope radius; at the end of a pass (third layer) it was 77 W at 25 turns a minute, over 75 W; the 75 W limit is the kit's constant, not a cited source | Design rate 23 turns a minute (D17): 71 W at the end of a pass, 64 W mean; duty cycle stated (about 45 %) |
| S4, S5, S6 | C45 normalised yields 305 MPa above 16 mm, not 340: the crank shaft fell to 1.89; the drum shaft check left out the gear's reaction | Both shafts C45+QT (430 MPa); crank shaft 30 mm through UCF206 bearings; gear reaction added to [C1] (159 MPa, margin 2.7); pinion Lewis at 305 MPa, margin 2.2 (D24) |
| S4 to S6, fatigue and contact | No fatigue or contact check; corroded machined shafts; pinion contact about 555 MPa at the rated pull | Crank shaft fatigue check added (S17, [C16]: 1.59 corroded); pinion teeth induction-hardened, contact check added (S28, [C3d]: 1.49) (D24) |
| S7 | The multi-layer factor of 1.8 is unsourced and the wall was chosen to pass; the margin of 2 holds only for a factor up to 1.95; flange thrust and welds not checked | S7 downgraded to Plausible, open until a sourced factor and a proof load with three layers wound on; flange thrust listed as open in REVIEW |
| S9 | Friction 0.30 is a dry lining; wet with brine (0.20) the hand force is 179 N | Lever 600 mm, design friction 0.20: 134 N; lining kept dry (D21) |
| S11, S12 | The chain and anchor checks assumed equal sharing; at 70 % on one chain the margins are 1.71 and 1.74; the anchor formula is for an axial pull | 10 mm chain, rated turnbuckles and shackles (D22), chain margin 2.6; anchor installed at 45 degrees so the load is 22 degrees off its axis; proof pull 3.42 kN; S12 stays open |
| S13 | Only the rated pull was checked; one slack chain would drop the margin below 1 | 6 kN case added ([C12b]: 1.46); a slack chain is a hazard controlled at set-up (RSK F11) |
| S31 | Static pressure only; the rope pulls above the drag, shifting load to the front of the skids | The pitching check (S29, [D13]) found the rake would tip onto its blade with the bridles at 130 mm; bridles lowered to 70 mm (D23); restoring moment now 9.8 times the pitching moment |
| PR-003 | "Every load-bearing part keeps a margin of 2" overclaimed: keys, pins, welds, plates, turnbuckles unchecked | Keys checked and fixed (S22, D24), roller axle checked ([C19]: 3.9); CAL now lists what is not calculated (plates, welds, turnbuckles) and leaves it to the proof load at TRL 4 |
| S16 | Arm cranking is nearer 15 % efficient: about 470 W metabolic, high work, reference WBGT about 26 C; this conflicts with 6 h a day | S16 text updated; S24 now gives the 4 h figure (about 130 t) beside the 6 h one; S16 stays open |
| S18 | Brine-wetted surfaces are CX, up to 25 um a year | Recomputed at 25 um a year: 2.2 years on the 3 mm runners, still more than a season |
| S20, S21, S27 | Rope abrasion on the crystal bed and fibre shedding not considered | Added to S21 and S27; S27 stays open; RSK F1 covers daily inspection |

| Row | Finding | What changed |
|---|---|---|
| Table 1 | Row numbers missing; the review table was empty | Gaps explained under Table 1; this table filled |

## Sources


- Business Standard (2009), "Growing pan of worries: livelihood of over 45k Agariyas in jeopardy": a family produces 500 to 600 t a year from one pan. https://www.business-standard.com/article/companies/growing-pan-of-worries-livelihood-of-over-45k-agariyas-in-jeopardy-109080500047_1.html
- EN 10083-2 (C45, normalised), EN 10025-2 (S275), IS 4923 (YSt 310 hollow sections), EN 10210 (S355J2H): yield strengths as stated in Table 1.
- EN 1993-1-8, Design of steel structures: design of joints (edge distance practice e1 >= 1.2 d0).
- ISO 1461, Hot dip galvanized coatings on fabricated iron and steel articles (minimum mean coating thickness).
- ISO 9223, Corrosion of metals and alloys: corrosivity of atmospheres (zinc corrosion rates by category).
- ISO 7243, Ergonomics of the thermal environment: assessment of heat stress using the WBGT index (reference values to confirm against the 2017 edition).
- Nickel Institute, guidance on stainless steels in seawater and chloride waters (crevice corrosion of type 316).
- ISO 6336, Calculation of load capacity of spur and helical gears (contact stress method and application factor, used in the manner of the standard).
- Grade 30 proof-coil chain working load limits (10 mm, 2,650 lb), catalogue class, to confirm on purchase.
- Snook, S. H., and V. M. Ciriello (1991), "The design of manual handling tasks: revised tables of maximum acceptable weights and forces," Ergonomics 34 (9): 1197 to 1213.
- RopeServices UK, 12 mm double braid polyester rope (breaking strength about 2,957 kg, less 10 % for each splice), checked 2026-10-25. https://www.ropeservicesuk.com/product/12mm-black-double-braid-polyester-rope-100-metre-reel/
- `.kit/stem_check.py`: sustained hand-cranking power of an average adult, 75 W.
- SDG-CAL-001, `docs/04-calcs/sizing.py` and `docs/04-calcs/results.csv`: every computed value in Table 1.